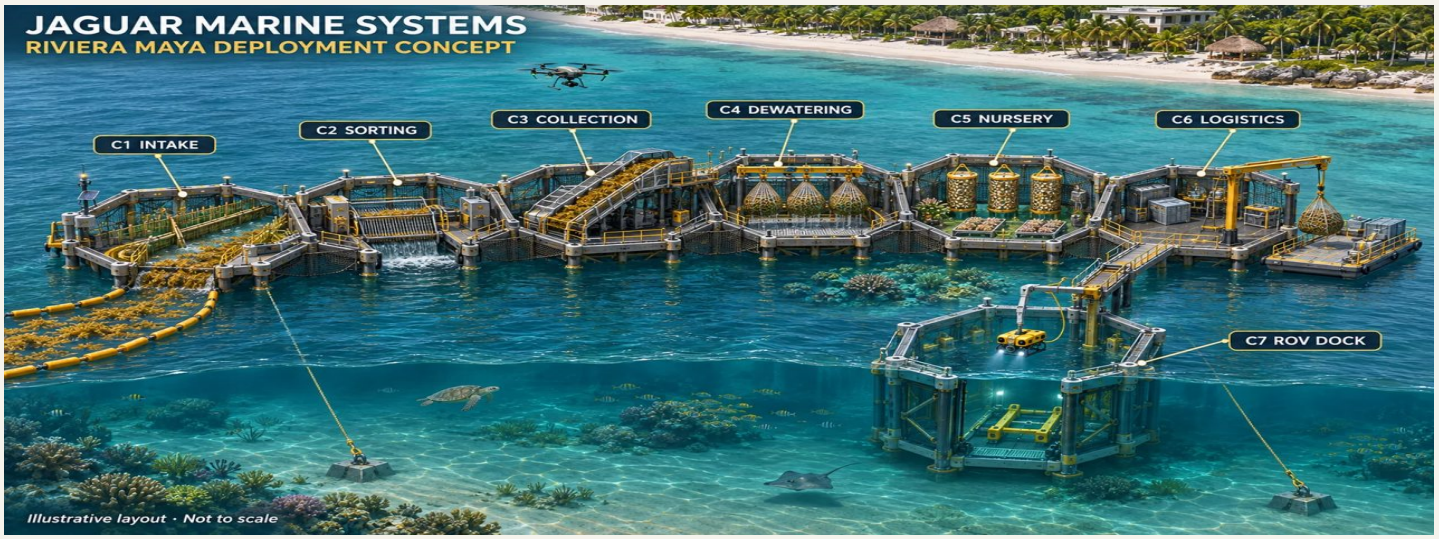
Prototype engineering brief

A proposed mechanical demonstrator for a modular coastal system



Riviera Maya deployment concept | Illustrative layout | Not to scale

PROPOSED FIRST BUILD

A dry bench article based on the common hex chassis (JH-210), with a representative habitat cartridge (JH-200), receiver (JH-205), and guide hardware. Final configuration and build scope require independent engineering review.

The objective is to establish the support and load conditions, resolve assembly fit, and demonstrate a documented service sequence. Later work packages address biomass handling, ROV docking, habitat trials, and marine deployment.

CURRENT EVIDENCE AND OPEN WORK

Cartridge CAD	JH-200 Rev H is recorded as a valid one-piece solid with integrated lifting eyes. Lift strength and buoyancy checks are open.
Support diagnostics	Direct-support numerical balance was reproduced. The physical bench supports, seating conditions, and contact reactions still need definition and review.
Service interface	Sampled CAD checks exist. Guide interference and findings from the staged extraction path require resolution.

> First gate Close LC-00 support and force/moment equilibrium for the agreed bench case before advancing to LC-01.

Plan, partners, and support

A staged work package with documented review decisions

01

Define and review the bench case

An independent engineer reviews the support surfaces, restraints, seating condition, load points, and acceptance limits. Demonstrate force and moment balance with credible support reactions.

02

Release the build and test package

Resolve interface clearances and extraction findings. Agree drawings, materials, fabrication scope, fixture design, handling provisions, and the test protocol before fabrication release.

03

Measure the demonstrator

Record assembly fit, service access, and the agreed load, displacement, and reaction measurements. Use accepted bench evidence to scope later controlled water trials.

PARTNERS SOUGHT

Marine structural and FEA review, CAD detailing, prototype fabrication, and instrumentation or test support. Coastal ecology input will help define the later habitat and water-trial work.

FUNDING SCOPE

Work package	Budget status
Engineering review and bench-case definition	Scope and quote required
CAD detailing, fixture, and fabrication	Quote after engineering review
Instrumentation and bench measurements	Quote against agreed test plan
Later controlled water trial	Separate phase to be scoped

Total funding and schedule will be set after scope review and supplier quotes. Support may include engineering hours, fabrication capacity, equipment, or sponsorship.

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Request the engineering review packet
Jaguar Marine Systems website

Basis - JMS development heatmap and LC-00 equilibrium audit, 10 Nov 2026. Rev B baseline retained. Detailed engineering evidence is available for reviewer handoff.